

LESSON 6-3 • SESSION 1 • PRACTICE • FORM A

Writing Repeated Factors as Powers

Learning target – I can write repeated multiplication as a power and name its base and exponent. • 6.EE.A.1

REMEMBER

- A **power** shows a repeated factor: $5 \times 5 \times 5 = 5^3$.
- The **base** is the repeated factor. The **exponent** tells how many times it is a factor.
- Read 5^3 as "5 to the third power" or "5 cubed." 5^2 is "5 squared."

WORKED EXAMPLEWrite $9 \times 9 \times 9 \times 9$ as a power.**Step 1** The repeated factor is 9, so the base is 9.**Step 2** Count the 9s: there are four, so the exponent is 4.**Step 3** Check: the exponent counts factors, not $\times$ signs (there are only 3 $\times$ signs).**Answer** 9^4 ; base 9, exponent 4**WORDS** **power** (*potencia*) • **base** (*base*) • **exponent** (*exponente*)**PRACTICE**

show your work • label units

1. Write each product as a power.

a. $7 \times 7 \times 7 \times 7 =$ _____

b. $4 \times 4 \times 4 \times 4 \times 4 =$ _____

c. $5 \times 5 \times 5 =$ _____

d. $8 \times 8 =$ _____

e. $3/4 \times 3/4 =$ _____

f. $0.4 \times 0.4 \times 0.4 =$ _____

- 2.
- Club and Patio.**
- After six weeks, a club's size has been multiplied by
- $2 \times 2 \times 2 \times 2 \times 2 \times 2$
- . A square patio has
- 10×10
- tiles.

- Write the club product as a power. _____
- Write the patio product as a power. _____
- In the club power, name the base and the exponent. _____
- What does the exponent count? _____

- 3.
- Game Rounds.**
- A score is multiplied by 5 in each of four rounds:
- $5 \times 5 \times 5 \times 5$
- . Write this as a power. Name the base and exponent.

Hint The repeated factor is the base.

Show your work.

Power _____

- 4.
- Ant Colony.**
- A science report records an ant count as
- $13 \times 13 \times 13$
- . Write this as a power. Name the base and exponent.

Hint Count the factors of 13.

Show your work.

Power _____

- 5.
- Block Tower.**
- Rafi wrote the product
- $8 \times 8 \times 8$
- as the power
- 8^8
- . Justify or correct his work.

Hint What does the exponent count? Recount the factors.**Power** _____

6. **Two Directions.** Write each one the other way.

Hint Put parentheses around a fraction base.

a. Write $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$ as a power. _____

b. Write 11^3 as a product of factors. _____

Show your work.

7. **Folding Paper.** Each time a sheet of paper is folded, the number of layers doubles. After 1 fold there are 2 layers.

a. Write the number of layers after 5 folds as a power of 2. _____

b. Find the number of layers after 5 folds. _____ layers

c. Explain how each new fold changes the power. _____

8. **Swapped Numbers.** Compare 4^3 and 3^4 .

a. Write each power in expanded form. _____

b. Which power has more factors? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Kayla's Work.** Kayla read the power 12^4 and wrote: " $12 \times 4 = 48$, so 12^4 equals 48." Explain her mistake. Then write 12^4 correctly as a product.

TEST PRACTICE

choose the best answer

10. $3 \times 3 \times 3 \times 3 \times 3$ written as a power is –

A 5^3

B 3^5

C 3×5

D 15

11. In 6^4 , the base is –

A 6

B 4

C 24

D 10

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

10 Z	2 ⁶ E	3 ⁴ T	2 ⁵ E	8 ³ O	32 N	13 ³ P	5 ⁴ X	8 ⁸ Q	(1/2) ³ N
#2a	#3	#4	#5	#6a	#7a	#7b	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____